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A Q-Theory of Banks

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1 Big picture

- **Question.** How does bank regulation work when regulatory capital is written on *book* values, but markets price banks using *forward-looking* information that recognizes losses faster than accounting books do?
- **Core mechanism.** The paper introduces **delayed loan-loss recognition**. Loan defaults reduce *fundamental* equity immediately, but *book* equity adjusts only gradually. The wedge is the stock of **zombie loans**, i.e. unrecognized losses that still sit on the books.
- **Why this matters.** Capital requirements are based on book values, so if losses are recognized slowly, regulation constrains banks using stale information. This can explain why market and book leverage look very different in crises.
- **Four motivating facts.** The paper documents that: (i) market equity and book equity diverge sharply in crises; (ii) Tobin's Q predicts future profitability and default risk; (iii) book-based capital requirements rarely bind tightly even while market leverage fans out in crises; and (iv) banks delever slowly after net-worth shocks.
- **Theory contribution.** The paper builds a dynamic bank model with three equity concepts: **fundamental equity** W , **book equity** $\bar{W}$, and **market equity** S . Market discipline acts on fundamental leverage, while regulators act on book leverage.
- **Main positive result.** Delayed accounting jointly explains the market-book wedge, the predictive content of Tobin's Q , the loose appearance of regulatory book constraints, and the slow post-shock deleveraging dynamics.
- **Main quantitative result.** In the estimated model, the loan-loss recognition speed is $\alpha = 4.34\%$, and the socially optimal reform combines **much faster** loss recognition ($\alpha \approx 23.6\%$) with a **modest relaxation** of the book capital requirement from roughly **8% to 5.4%**.
- **Policy message.** Accounting rules are not a sideshow to bank regulation. They are an additional policy instrument. Faster accounting recognition can lower hidden risk-taking and allow regulators to use less stringent book capital ratios.
- **Macroprudential message.** A countercyclical capital buffer (CCyB) can backfire under delayed accounting because it relaxes constraints for all banks, including those not yet hit by losses. A relaxation of accounting rules is better targeted because it primarily helps banks that have actually been hit by shocks.

2 Data and measurement

2.1 Bank panel and sample construction

- **Sample.** Publicly traded U.S. bank holding companies (BHCs), 1990Q3–2021Q1.
- **Regulatory accounting data.** FR Y-9C reports provide balance-sheet and income-statement information.
- **Market data.** CRSP provides stock prices and shares outstanding.
- **Auxiliary calibration targets.** FDIC bank failure data and aggregate charge-off information are used in estimation and in aggregate-crisis exercises.

2.2 Key empirical objects

- **Tobin's Q .** The paper defines Tobin's Q as the *market-to-book equity ratio*, not the market-to-book asset ratio:

$$Q = \frac{\text{Market Equity}}{\text{Book Equity}}.$$

- **Book equity measures.** The main book measures are total book equity and Tier 1 capital.
- **Market leverage.** Empirically, market leverage is constructed from liabilities and market capitalization; in the model, the natural analogue is assets over market equity or fundamental equity.
- **Return on equity.** Future ROE is next year's net income divided by current book equity.
- **Distance to default.** The paper uses a Z-score-style distance-to-default object built from market equity, book debt, and the volatility of bank value growth.
- **Net-worth shocks.** These are measured from risk-adjusted excess stock-return shocks at the bank-quarter level.

2.3 Model objects mapped to the data

- **Fundamental equity W .** The true economic equity value once losses are immediately recognized.
- **Book equity $\bar{W}$.** The accounting measure used by regulators.
- **Market equity S .** Equity valued by outside shareholders.
- **Zombie loans Z .** Unrecognized losses that stay on the books as loans.
- **Zombie-loan ratio.**

$$z = \frac{Z}{\bar{W}}.$$

- **Fundamental leverage.**

$$\lambda = \frac{L}{W}.$$

- **Book leverage.**

$$\bar{\lambda} = \frac{\bar{L}}{\bar{W}} = \frac{\lambda + z}{1 + z}.$$

A larger z makes the bank look less levered on its books than it really is.

2.4 Estimated parameter values and the moments behind them

- **Externally set parameters.** The loan yield is $r^L = 1.01\%$, the bank debt yield is $r^D = 0.51\%$, and the regulatory book leverage limit is $\Xi = 12.5$, which corresponds to an 8% capital requirement.
- **Estimated parameters.** The SMM exercise estimates

$$\rho = 2.02\%, \quad \rho^I = 3.45\%, \quad \theta = 6.69, \quad \varepsilon = 1.24\%, \quad \alpha = 4.34\%, \quad \kappa = 50.37, \quad \sigma = 0.105, \quad v_0 = 0.037.$$

- **Moment mapping.**

- ρ : book equity growth rate (2%).
- ρ^I : average market-to-book ratio (1.316).
- θ : IRF of market leverage.
- ε : average book leverage (11.36).
- α : mean reversion in liabilities and leverage IRFs.
- κ : initial liabilities response and mass of banks exposed to market liquidation.
- σ : mean quarterly net charge-off rate (0.12%).
- v_0 : quarterly bank failure rate (3.65 basis points).

3 Models and empirical specifications

3.1 Reduced-form evidence: distributed-lag impulse responses

The paper estimates the dynamics of bank variables after a bank-specific net-worth shock using

$$\Delta \log(y_{i,t}) = \alpha_t + \sum_{h=0}^k \beta_h \varepsilon_{i,t-h} + \psi_{i,t}. \quad (1)$$

- **Interpretation.** $y_{i,t}$ is a bank outcome such as Tobin's Q , market leverage, market equity, or liabilities.
- **Shock.** $\varepsilon_{i,t}$ is an idiosyncratic net-worth shock constructed from bank excess returns after purging common factor movements.
- **Time fixed effects.** α_t purge aggregate shocks, so the estimated IRFs are partial-equilibrium cross-sectional responses.
- **Impulse response.** The cumulative response at horizon k is $\sum_{h=0}^k \beta_h$.

3.2 Bank environment and the law of motion for equity

Banks fund risky loans with deposits and internal equity. The bank solves

$$V_0 = \mathbb{E} \left[\int_0^\infty e^{-\rho t} C_t dt \right], \quad (1)$$

where dividends satisfy $C_t = cW_t$ in the baseline theoretical exposition.

At each instant the bank chooses leverage $\lambda \geq 1$ with

$$L = \lambda W, \quad D = (\lambda - 1)W.$$

Fundamental equity follows

$$dW = [r^L \lambda - r^D (\lambda - 1) - c] W dt - \varepsilon \lambda W dN. \quad (2)$$

- **Economic meaning.** Leverage scales both the return on equity and the size of losses when defaults occur.
- **Default process.** dN is a Poisson process with intensity σ ; when it jumps, a fraction ε of loans defaults.
- **Key implication.** Losses mechanically increase leverage after a shock, even before the bank adjusts its balance sheet.

3.3 Book equity, fundamental equity, market equity, and Tobin's Q

Delayed accounting creates a wedge between fundamental and book values. Book loans and book equity are

$$\bar{L} \equiv L + Z, \quad \bar{W} \equiv \bar{L} - D = W + Z.$$

Define the zombie-loan ratio as

$$z \equiv \frac{Z}{W}, \quad \bar{\lambda} = \frac{\lambda + z}{1 + z}.$$

The paper decomposes Tobin's Q as

$$Q \equiv \frac{S}{\bar{W}} = \frac{S}{W} \cdot q, \quad q \equiv \frac{W}{\bar{W}} = \frac{1}{1 + z}. \quad (3)$$

- **Interpretation of q .** The paper's distinctive object is the fundamental-to-book equity ratio. When zombie loans are large, q is small, so market-to-book values look depressed even absent changes in shareholder discount rates.
- **Interpretation of S/W .** This captures the valuation wedge between shareholders and bankers due to different discount rates.
- **Takeaway.** The paper's Tobin's Q is driven both by hidden losses in books and by valuation differences in market prices, but the hidden-loss margin is the central new mechanism.

3.4 Liquidation constraints, delayed recognition, and survival boundaries

Banks can be liquidated either by market discipline or by regulators. The two primitive constraints are

$$\frac{L}{W} \leq \kappa, \quad \frac{\bar{L}}{\bar{W}} \leq \Xi. \quad (4)$$

These can be combined into a single liquidation boundary in (λ, z) space:

$$\lambda \leq \Gamma(z) \equiv \min\{\kappa, \Xi + (\Xi - 1)z\}. \quad (5)$$

After a default, the market-liquidation condition is

$$\lambda + J^\lambda = \frac{\lambda(1 - \varepsilon)}{1 - \varepsilon\lambda} \leq \kappa, \quad (6)$$

and the regulatory-liquidation condition is

$$\bar{\lambda} + J^{\bar{\lambda}} = \frac{\lambda(1 - \varepsilon) + z}{1 - \varepsilon\lambda + z} \leq \Xi. \quad (7)$$

The **shadow boundary** is the highest leverage such that the bank survives a default shock:

$$\Lambda(z) = \min \left\{ \frac{\Xi + (\Xi - 1)z}{1 + (\Xi - 1)\varepsilon}, \frac{\kappa}{1 + (\kappa - 1)\varepsilon} \right\}. \quad (2)$$

When z is high enough, the market constraint dominates and $\Lambda(z)$ becomes flat.

- **Main intuition.** More hidden losses make the bank *look* safer on the books, so the book constraint relaxes in terms of fundamental leverage.
- **Why the distinction matters.** The bank can be safe in accounting space while remaining fragile in fundamental or market-value space.
- **Core dynamic object.** The gap between the shadow boundary $\Lambda(z)$ and the liquidation boundary $\Gamma(z)$ determines whether banks immediately delever after a shock or choose to sit near closure.

3.5 Zombie-loan dynamics

Zombie loans evolve as

$$dZ = -\alpha Z dt + \varepsilon \lambda W dN, \quad (8)$$

and therefore the zombie ratio evolves according to

$$dz = -z(\alpha + \mu^W) dt + \lambda \varepsilon \left[\frac{1 + z}{1 - \lambda \varepsilon} \right] dN. \quad (9)$$

- **Interpretation of α .** This is the speed of loan-loss recognition. It is the paper's main accounting-policy parameter.
- **Economic meaning.** Defaults create hidden losses instantly after a bank survives the shock, but accounting books only recognize those losses gradually over time.
- **Connection to the data.** A lower α generates persistent wedges between market and book values and slow leverage adjustment, exactly as in the empirical facts.

3.6 The bank's dynamic program

The state variables are (Z, W) , and the bank chooses leverage λ to solve

$$\begin{aligned} \rho V(Z, W) = & \max_{\lambda \in [1, \Gamma(Z/W)]} \left\{ cW + V_Z(Z, W)\mu^Z W + V_W(Z, W)\mu^W W \right. \\ & \left. + \sigma [V(Z + J^Z, W + J^W)\mathbf{1}_{\lambda \leq \Lambda(Z/W)} + v_0 W \mathbf{1}_{\lambda > \Lambda(Z/W)} - V(Z, W)] \right\}. \end{aligned} \quad (10)$$

- **Interpretation.** The bank trades off levered profits against the value destruction from crossing survival boundaries after a default.
- **Continuation value.** If the bank survives a default, it continues with new state $(Z + J^Z, W + J^W)$.
- **Liquidation value.** If the bank fails, bankers recover $v_0 W$.
- **Market value of equity.** Shareholders solve the same HJB with discount rate ρ^I and are wiped out upon liquidation.

3.7 Immediate accounting benchmark

If accounting is immediate, then $\alpha \rightarrow \infty$ and $z_t = 0$. The bank's value is linear in equity,

$$V(0, W) = vW,$$

and the optimal expected levered return is

$$\Omega^* = r^D + \max_{\lambda \in [1, \kappa]} \left\{ (r^L - r^D)\lambda + \sigma \left[(1 - \varepsilon\lambda)\mathbf{1}_{\lambda \leq \Lambda} + \frac{v_0}{v}\mathbf{1}_{\lambda > \Lambda} - 1 \right] \right\}. \quad (11)$$

The threshold between staying safe and risking liquidation is given by λ^o , which solves

$$(r^L - r^D) \left(\lambda^o - \frac{\lambda^o}{1 + \varepsilon(\lambda^o - 1)} \right) = \sigma \left(1 - \frac{v_0}{v} - \varepsilon \frac{\lambda^o}{1 + \varepsilon(\lambda^o - 1)} \right). \quad (12)$$

With a regulatory cap, optimal leverage is

$$\lambda^*(\Xi, \kappa) = \begin{cases} \min\{\kappa, \Xi\} (1 + \varepsilon(\min\{\kappa, \Xi\} - 1))^{-1}, & \text{if } \min\{\kappa, \Xi\} \leq \lambda^o, \\ \min\{\kappa, \Xi\}, & \text{if } \min\{\kappa, \Xi\} > \lambda^o. \end{cases} \quad (13)$$

- **Bang-bang logic.** The bank either sits at the shadow boundary (safe) or at the liquidation boundary (risky). There is no interior optimum.
- **Why immediate accounting fails empirically.** It implies no internal propagation, a constant Tobin's Q , and no predictive role for Q once book values are controlled for.

3.8 Delayed accounting solution

With delayed accounting, value remains homogeneous in W but depends on the zombie ratio z :

$$V(Z, W) = v(z)W, \quad L(Z, W) = \lambda^*(z)W.$$

The value function satisfies

$$\rho v(z) = c - v_z(z)\alpha z + (v(z) - v_z(z)z) [\Omega^*(z) - c], \quad (14)$$

with

$$\Omega^*(z) = r^D + \max_{\lambda \in [1, \Gamma(z)]} \left\{ (r^L - r^D)\lambda + \sigma \frac{J^v(z, \lambda)}{v(z) - v_z(z)z} \right\},$$

where

$$J^v(z, \lambda) = v(z + J^z)(1 - \varepsilon\lambda)\mathbf{1}_{\lambda \leq \Lambda(z)} + v_0\mathbf{1}_{\lambda > \Lambda(z)} - v(z).$$

Optimal leverage is again bang-bang:

$$\lambda^*(z) = \begin{cases} \Lambda(z), & \text{if } \Omega(z, \Gamma(z)) \leq \Omega(z, \Lambda(z)), \\ \Gamma(z), & \text{if } \Omega(z, \Gamma(z)) > \Omega(z, \Lambda(z)). \end{cases} \quad (15)$$

- **Key state variable.** z determines whether the bank wants to stay safely at the shadow boundary or take the gamble of remaining at the liquidation boundary.
- **Figure 6 intuition.** After a default, a bank with moderate z immediately delevers back to the shadow boundary, but a bank with very high z may remain on the liquidation boundary until enough hidden losses are recognized.
- **Why this matters quantitatively.** This is the source of slow leverage dynamics and book-market wedges in the model.

3.9 Social welfare and the policy problem

The social planner values aggregate discounted bank payouts starting from an initial distribution $g_0(z, W)$:

$$P(\alpha, \Xi, \{g_0\}) = \int_0^\infty \int_0^\infty \mathbb{E} \left[\int_0^\infty e^{-\rho t} c W_t dt \mid W_0 = W, z_0 = z \right] g_0(z, W) dz dW. \quad (16)$$

The planner assumes liquidation is socially costly:

$$r^L - r^D \leq \sigma(\varepsilon + (1 - \psi)(1 - \varepsilon)).$$

Under immediate accounting, the first-best leverage solves

$$\Pi^{\text{fb}} = \max_{\lambda} \left(r^L - r^D - \sigma[\varepsilon + (1 - \psi)(1 - \varepsilon)\mathbf{1}_{\lambda > \Lambda}] \right) \lambda, \quad (17)$$

and second-best capital regulation solves

$$\Pi^{\text{sb}} = \max_{\Xi} \left(r^L - r^D - \sigma[\varepsilon + (1 - \psi)(1 - \varepsilon)\mathbf{1}_{\lambda^*(\Xi, \kappa) > \Lambda}] \right) \lambda^*(\Xi, \kappa). \quad (18)$$

With delayed accounting, the social value can be written as

$$P^*(\{g_0\}) = \max_{\{\alpha, \Xi\}} \int_0^\infty \int_0^\infty W p(z) g_0(z, W) dz dW, \quad (19)$$

where $p(z)$ satisfies a planner's HJB that differs from the bank's private HJB only because the planner internalizes liquidation losses.

- **Central normative insight.** Capital requirements are second-best because enforcing them requires liquidation, which is itself socially costly.
- **Why α matters.** Changing accounting speed alters the cross-sectional distribution of hidden losses and thus changes the amount of regulatory buffer banks choose to hold.
- **Welfare approximation used in the quantitative discussion.**

$$P(\alpha, \Xi, \{g_0\}) \approx \frac{C - S}{\rho - (G - C)},$$

where C is the aggregate dividend rate, S the flow of social losses from liquidation, and G the aggregate ROE before dividends.

3.10 Macroprudential policy experiments

To study macroprudential policy, the paper introduces a boom-bust path for default intensity:

$$\sigma_t = \sigma \left(1 - \frac{1}{2} e^{-\eta(t-\tau)^2} \right). \quad (20)$$

It then compares a benchmark regime to a CCyB regime with time-varying capital requirements:

$$\Xi_t = \Xi \left(1 + \left(\frac{\Xi^{\text{CCyB}}}{\Xi} - 1 \right) e^{-\eta(t-\tau)^2} \right), \quad (21)$$

and to a time-varying accounting-recognition rule α_t^{CCyB} with the same shape.

- **Interpretation.** The CCyB relaxes capital constraints for all banks during the crisis; counter-cyclical accounting instead relaxes loss recognition mainly for banks that have already been hit by shocks.
- **Main result.** The CCyB can increase long-run fragility by encouraging more ex ante risk-taking, whereas temporary slower recognition of losses is better targeted and produces fewer failures.

4 Identification and what makes the estimates credible

4.1 Identification of the empirical shock

- **Main challenge.** A bank stock return mixes cash-flow news, risk-premium movements, and changes in investment opportunities. The paper wants only the *net-worth shock* component.
- **Solution.** Each bank's excess return is decomposed into common-factor and idiosyncratic parts using a **five-factor model** (the three Fama-French factors plus credit and interest-rate factors).
- **Interpretation.** The residualized excess return is treated as a risk-adjusted, bank-specific shock to net worth, analogous to the loan-default shock in the structural model.
- **Why this is credible.** Under the efficient-markets logic used by the paper, these residual excess returns are the cleanest observable proxy for unexpected shocks to the bank's franchise value and cash flows.

4.2 Identification of structural parameters

- ρ . Identified by average book-equity growth because the banker's discount rate governs payout behavior.
- ρ^I . Identified by average market-to-book ratios because the shareholder discount rate governs S/W .
- θ . Identified by the response of market leverage because dividend smoothing affects the speed and composition of recapitalization.
- ε . Identified by mean book leverage through the distance between the shadow and liquidation boundaries.
- α . Identified by the slow mean reversion in liabilities and leverage IRFs. A slower recognition rate implies slower deleveraging.
- κ . Identified by how many banks are effectively exposed to market liquidation risk, which shows up in the initial liabilities response.
- σ and v_0 . Pinned down by average charge-off rates and observed bank failure rates.

4.3 Why delayed accounting is not just a reduced-form adjustment cost

- Immediate accounting with balance-sheet adjustment costs could, in principle, generate slow deleveraging.

- But it would **not** generate the fact that Tobin's Q predicts future ROE and future charge-offs after controlling for book equity, because all losses would already be on the books.
- The delayed-accounting mechanism therefore does two things at once: it generates inertia *and* informational content in the wedge between market and book values.

4.4 Key maintained assumptions

- **Regulators cannot liquidate a bank using market values alone.** They must intervene and build accounting-based evidence.
- **Banks cannot hide losses instantaneously.** They can eventually roll losses into zombie loans, but there is a brief window around the default event when a regulator could intervene.
- **Deposit pricing does not discipline leverage.** Deposits are effectively risk-free because of deposit insurance or guarantees, so banks do not privately internalize full liquidation costs.
- **Why these assumptions matter.** Without them, book regulation would either be irrelevant or there would be no room for delayed accounting to matter quantitatively.

4.5 Aggregate-shock implementation for the crisis exercises

- To match the global financial crisis time series, the paper feeds in temporary aggregate shocks to σ , α , and v_0 .
- Specifically, the loan-default intensity rises from 0.105 to 0.516 from 2007Q3 to 2009Q4; the recognition speed rises from 4.34% to 10.09% through 2019Q4; and the liquidation value falls from 0.037 to 0.01 over the same post-2007 interval.
- These shocks let the model match the charge-off path, the cumulative bank-failure rate, the decline in book loans, and the crisis-time behavior of Tobin's Q .

5 Tables and figures

5.1 Figure 1: Tobin's Q and aggregate bank equity

Read:

- Panel (a) shows that aggregate bank Tobin's Q swings sharply over time, especially in crises.
- Panel (b) is the key visual fact: during the GFC, market equity collapses while book equity remains relatively smooth.
- This is the empirical motivation for distinguishing market from book measures in the theory.

5.2 Figure 2: Tobin's Q predicts future ROE and distance to default

Read:

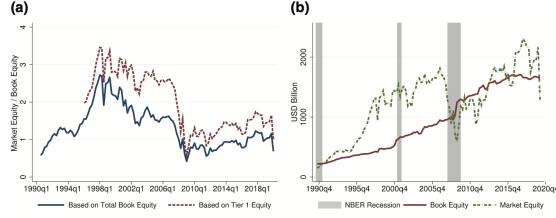


FIGURE 1
Tobin's Q and bank equity evolution (a) Tobin's Q and (b) Book and market equity
Notes: These figures show data on Tobin's Q in (a) and book equity and market equity in (b) for an aggregate sample of publicly traded BHCs. Tobin's Q is the ratio of market equity to book equity and the ratio of market equity to Tier 1 equity capital (only available since 1996). Book equity and Tier 1 equity are from the FR Y-9C. Market equity is from CRSP. Market equity equals shares outstanding times the share price, aggregated across publicly traded BHCs. All level variables are converted to 2012 Q1 dollars with the seasonally adjusted GDP deflator.

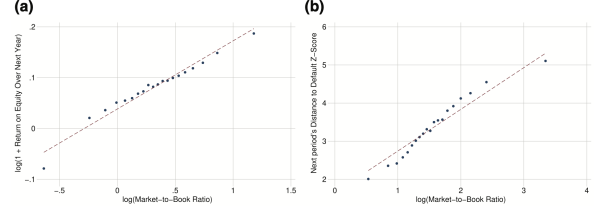


FIGURE 2
More cash flow-relevant information in market than in book equity (a) Return on equity and (b) Distance to default
Notes: This figure presents cross-sectional binned scatter plots of log outcomes on the log Tobin's Q for BHCs. All plots include controls for log book equity, the Tier 1 capital ratio, and quarter fixed effects. Data on market equity are from CRSP. All other data are from the FR Y-9C reports. Return on equity over the next year is defined as book net income over the next four quarters divided by book equity in the current quarter. The Z-score distance-to-default measure over one quarter is calculated at the bank level as $\frac{\log(V/D) + \mu_V - \frac{1}{2}\sigma_V^2}{\sigma_V}$ (see Duffie 2022). V denotes the total value of the bank measured as sum of the market value of equity and the book value of debt. D is measured as the book value of debt, or total liabilities. μ_A is the quarterly growth rate of V . σ_V is the standard deviation of the growth rate of V .

- The upward-sloping relationship says that higher market-to-book ratios predict higher next-year profitability and a larger distance to default.
- Book values do not fully contain the relevant information, even after controlling for book equity and capitalization.
- This is direct evidence in favor of the paper's hidden-loss interpretation of Q .

5.3 Figure 3: Regulatory constraints and market leverage quantiles

Read:

- Panel (a) shows that very few banks sit right at the regulatory capital boundary, even in the GFC; most hold capital buffers.
- Panel (b) shows that market leverage dispersion explodes in crises rather than compressing.
- Together, these patterns reject a model where market or book leverage constraints bind tightly for most banks at all times.

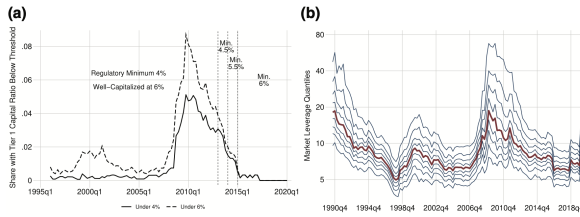


FIGURE 3
Leverage constraints (a) Regulatory constraints and (b) Market leverage quantiles
Notes: This figure shows the distribution of bank holding companies constrained by capital requirements in (a) and the quantiles of market leverage in Panel (b) for BHCs on a log scale. (a) The share of banks whose regulatory Tier 1 capital ratio, defined as (Tier 1 Capital)/(Risk-Weighted Assets), falls below a given threshold, computed from the full, unweighted sample is plotted. The regulatory capital requirements are shown on the graph and described in Supplementary Appendix A.2.1. Book data (liabilities) come from the FR Y-9C, and market equity data are from CRSP. In (b), market leverage is computed as (Liabilities + Market Equity)/Market Equity. The median value is plotted in bold red. Each tenth percentile is plotted in the thin blue lines.

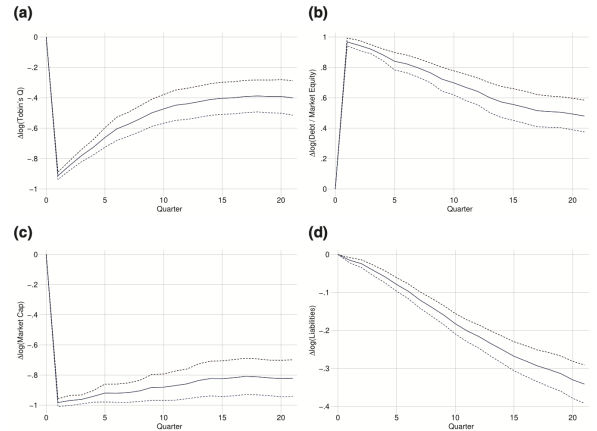


FIGURE 4
Estimated impulse responses (a) Tobin's Q, (b) Market leverage, (c) Market equity and (d) Liabilities
Notes: These figures show the estimated percent impulse responses to a 1% negative return shock. The y-axis of our plots shows the contemporaneous response ($-\beta_0$) as quarter 1, the cumulative response one quarter later ($-\beta_0 - \beta_1$) as quarter 2, and so on. Dashed lines denote the 95% confidence interval. Standard errors are clustered by bank. The panels display the impulse responses of log Tobin's Q in Panel (a), log market leverage in Panel (b), log market equity in Panel (c), and log liabilities in Panel (d). Market leverage is defined as (Liabilities/Market Capitalization). The sample is publicly traded BHCs from 1990Q3 to 2021Q1 using FR-Y-9C and CRSP data.

5.4 Figure 4: Empirical impulse responses to a negative net-worth shock

Read:

- Tobin's Q drops immediately and recovers only gradually.
- Market leverage jumps on impact and then mean-reverts slowly.
- Liabilities fall persistently, while market equity barely recovers.
- The overall message is that banks delever slowly and mostly through shrinking liabilities rather than quickly rebuilding equity.

5.5 Figure 5: Fundamental versus accounting balance sheets

Read:

- This is the clean balance-sheet picture behind the entire paper.
- Fundamental balance sheets carry only true assets; accounting balance sheets add zombie loans to both assets and equity.
- The figure makes clear why a bank can look well capitalized in accounting terms while being fragile in economic terms.

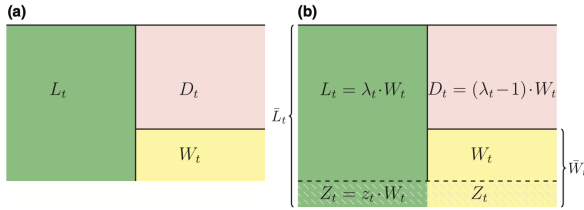


FIGURE 5
Fundamental and accounting balance sheet (a) Fundamentals and (b) Accounting

Notes: This figure shows the balance sheet of the bank in terms of fundamental values (a) and book values (b).

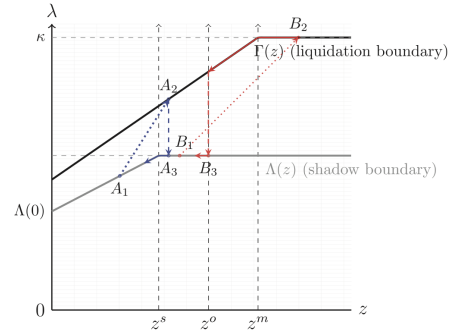


FIGURE 6
Typical trajectories of z and λ under delayed accounting

Notes: These panels depict the shadow and liquidation boundaries and typical trajectories under delayed accounting, as characterized in Proposition 2. The dashed vertical lines correspond to z^l (the point at which the market constraint becomes binding for the shadow boundary), z^o (the level of the zombie loan ratio above which a bank chooses to stay at the liquidation boundary and risk closure), and z^m (the point at which the market constraint becomes binding for the liquidation boundary). The blue trajectory shows the path of a bank starting at A_1 after it is hit with a shock; because the shock does not push the bank past z^o , it immediately delevers back to the shadow boundary. The red trajectory shows the path of a bank starting at B_1 ; after it is hit with a shock, it stays at the liquidation boundary and only returns to the shadow boundary once its zombie loan ratio drifts down to z^o .

5.6 Figure 6: Shadow and liquidation boundaries under delayed accounting

Read:

- This figure is the best way to understand the model's dynamics.
- A bank hit by a moderate shock jumps to a higher z and delevers immediately back to the shadow boundary.
- A bank hit by a severe enough shock can choose to remain on the liquidation boundary and risk closure until enough hidden losses are recognized.
- The discontinuity at z^o is what makes optimal leverage a bang-bang control.

5.7 Table 1: Parameterization

Read:

- The table maps each estimated parameter to a specific targeted moment.
- The two most economically informative parameters are $\alpha = 4.34\%$ and $\kappa = 50.37$: the first governs recognition speed, and the second determines how far away the market-based liquidation boundary is.
- The table is important because it shows the paper is not simply calibrating accounting delay arbitrarily; it is tying it to observed IRF persistence.

TABLE 1
Parametrization

Parameter	Description	Target
<i>Externally set parameters</i>		
$r^L = 1.01\%$	Loan yield	BHC data: avg. interest income/loans
$r^D = 0.51\%$	Bank debt yield	BHC data: avg. interest expense/debt
$\Xi = 12.5$	Regulatory maximum asset to equity ratio	Capital requirement of 8%
<i>Jointly determined—estimated</i>		
$\rho = 2.02\%$	Banker's discount rate	Book equity growth rate: 2%
$\rho^I = 3.45\%$	Investor's discount rate	Market-to-book ratio of equity: 1.316
$\theta = 6.69$	Banker's inverse IES	Market leverage IRF
$\epsilon = 1.24\%$	Loan loss rate in event of default	Mean book leverage
$\alpha = 4.34\%$	Speed of loan loss recognition	Liabilities IRF
$\kappa = 50.37$	Market-based leverage constraint	Mean quarterly net charge-off rate of 0.12%
$\sigma = 0.105$	Arrival rate of loan default shocks	Quarterly bank failure rate of 3.65 basis points
$v_O = 0.037$	Bank liquidation value	

Notes: This table summarizes the parameter values, their role in the model, and the data target used to set or estimate their value. The text provides more details.

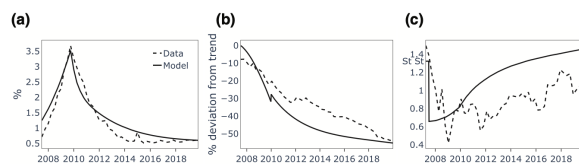


FIGURE 7
Crisis and recovery in model and data (a) Net charge-off rate, (b) Loan book values and (c) Tobin's Q (Fact 1)

Notes: This figure compares the aggregate series of model-generated data (solid line) to the empirical data (dashed line). These series are based on a simulation that feeds in shocks chosen to match the times series of aggregate net charge-off rates (Panel (a)). Panel (b) presents the evolution of book loans as a deviation from a trend based on the 10 years before 2006 Q4. Panel (c) presents Tobin's Q . We plot the steady-state value of Tobin's Q for 2007 Q2—before the aggregate shock is realized—as a reference. The labels on the x-axis refer to the first quarter of the year.

5.8 Figure 7: Crisis and recovery in the model and in the data

Read:

- Panel (a) shows that the imposed aggregate shock path matches the net charge-off rate quite closely.
- Panel (b) shows that the model captures the large decline in book loans after the crisis.
- Panel (c) shows that delayed accounting generates the sharp fall in Tobin's Q during the GFC.
- The fit is not just qualitative; it captures the basic time-series magnitudes of the crisis episode.

5.9 Figure 8: Model-implied predictability

Read:

- The model reproduces the cross-sectional positive slope between Tobin's Q and future ROE.
- It also reproduces the positive link between Tobin's Q and distance to default.
- This is important because it shows the model does not merely fit time-series wedges; it also fits the cross-sectional information content of market valuations.

5.10 Figure 9: Crisis-time dispersion in market leverage

Read:

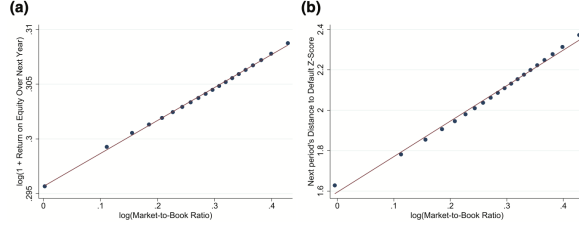


FIGURE 8

Predictability (Fact 2) (a) Return on equity and (b) Distance to default

Notes: (a) A cross-sectional binscatter plot of next-year's ROE over log Tobin's Q (log market-to-book ratio of equity) is presented. (b) A cross-sectional binscatter plot of the distance to default over log Tobin's Q (log market-to-book ratio of equity) is presented. Both figures control for the book value of equity and equity capitalization. The data are from model-simulated data using the stationary distribution and the parametrization in Table 1.

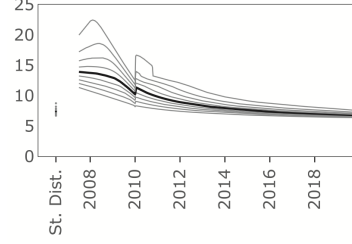


FIGURE 9

Market leverage dispersion (Fact 3)

Notes: This figure shows the distribution of market leverage for model-simulated data in response to the same aggregate shocks as in Figure 7. The bold line is the median, and the thin lines are cross-sectional deciles of market leverage. The stationary distribution of market leverage is plotted for reference. The year labels on the x -axis refer to the first quarter of the year.

- The model generates a widening cross-sectional distribution of market leverage during the GFC.
- Quantitatively, the paper notes that it captures about one-third of the increase in dispersion.
- This is enough to validate the mechanism, while leaving room for omitted forces such as ex ante heterogeneity or time-varying risk premia.

5.11 Figure 10: Model versus data impulse responses

Read:

- The model produces the slow recovery in Tobin's Q , the gradual deleveraging, and the persistent decline in liabilities.
- The weakest fit is on market equity recovery, but even there the model gets the broad fact right that recapitalization is extremely slow.
- This figure is the central quantitative validation of the delayed-accounting mechanism.

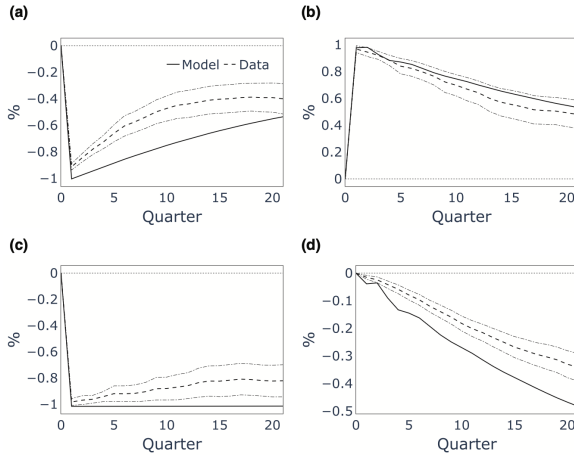


FIGURE 10

Model versus data impulse responses (Fact 4) (a) Tobin's Q , (b) Market leverage, (c) Market equity and (d) Liabilities

Notes: The figures present the impulse response functions of model-simulated data (solid line) for the benchmark calibration and compares them to those from the data (the dashed line represents the point estimates and the dash-dot lines the 95% confidence interval). We show the impulse response function of Tobin's Q in (a), market leverage in (b), market equity in (c), and liabilities in (d). To compute the model IRFs, we first solve and simulate the model using the baseline parameter values from Table 1 and construct bank market returns as explained in [Supplementary Section G.2 equation \(72\)](#). We run pooled OLS regressions of the demeaned variable of interest on banks' market return and 20 lags using the simulated data. Finally, we take the coefficients, multiply each by -1% , and compute the IRF at horizon h as the sum of the coefficients up to lag h .

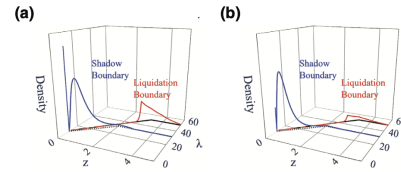


FIGURE 11

Comparison of stationary distributions for different α (a) Baseline parameters: $\alpha = 4.34\%$ and (b) Faster recognition: $\alpha = 6\%$

Notes: This figure presents a two-dimensional plot of the stationary distribution of banks across fundamental leverage λ and the zombie loan ratio z . The black dashed line traces out the shadow boundary $\Lambda(z)$ and the solid black line the liquidation boundary $\Gamma(z)$. The blue and red lines are the density of banks conditional on their choosing the shadow and liquidation boundaries, respectively. For visualization purposes, the density conditional on banks' choosing the liquidation boundary has been multiplied by 20, as it is otherwise not visible. (a) Sets parameters at their benchmark levels in Table 1, and (b) increases α to 6%.

5.12 Figure 11: Stationary distributions for different loss-recognition speeds

Read:

- Raising α shifts the distribution of banks toward lower z and lower fundamental leverage.
- Fewer banks sit on the liquidation boundary when recognition is faster.
- The figure provides the distributional reason that faster accounting can reduce failures.

5.13 Figure 12: The tradeoff induced by α

Read:

- Panel (a) shows a strong negative relationship between the zombie ratio and fundamental leverage as α rises.
- Strikingly, average *book* leverage barely moves even as fundamental leverage changes a lot. So book statistics alone can be misleading about true fragility.
- Panel (b) shows the core tradeoff: faster recognition reduces both loan growth and bank failure over a broad range.

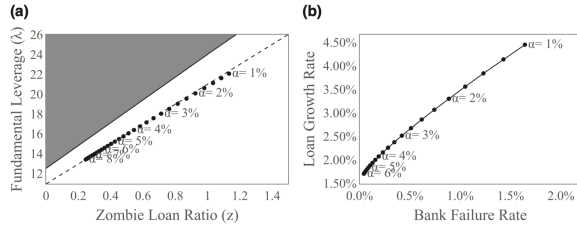


FIGURE 12

Steady states for different α : effect of delayed loss recognition on λ and z (a) Leverage versus zombie loan ratios and (b) Liquidations versus loan growth

Notes: (a) Cross-sectional averages of λ and z from the stationary distribution for different values of α is presented. The grey shaded area presents the liquidation set and the dashed line the shadow boundary. (b) The cross-sectional average of banks' loan growth rate and the average bank failure rate for different levels of α is presented. We assume that all other parameters remain at their benchmark levels shown in Table 1.

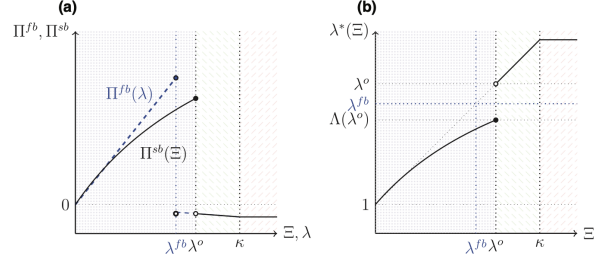


FIGURE 13

First-best leverage and optimal regulation (immediate accounting) (a) Planner objectives and (b) Best-response leverage
Notes: These panels show the regulator's problem under immediate accounting, in terms of the planner's objectives (left panel) and the bank's behaviour λ^* in response to regulation (right panel) for the case where $\kappa > \lambda^o$. The planner's objective under the first best is denoted by Π^{fb} and under the second best by Π^{sb} . The right panel shows the bank's best response.

5.14 Figure 13: First-best leverage and second-best regulation under immediate accounting

Read:

- The first-best planner would like leverage at the market shadow boundary: as high as possible without triggering liquidation risk.
- Capital regulation cannot implement that first best because banks respond by holding buffers.
- This is the conceptual reason the paper needs a second regulatory instrument beyond book capital requirements.

5.15 Figure 14: Optimal microprudential policy

Read:

- The welfare peak lies at *high* α and *lower* capital requirements than in the baseline.
- Relative to the estimated point, the social optimum is around $\alpha = 23.6\%$ and a capital requirement near 5.4% .
- The reform lowers annualized failure costs substantially and reduces the hidden-risk problem, even though it allows banks to look more levered on the books.

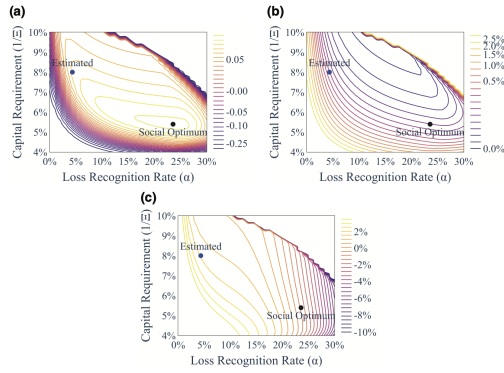


FIGURE 14
Optimal microprudential policy and isovalues (a) Welfare, (b) Bank failure rates and (c) loan growth rates
Notes: (a) The values of social welfare $\mathcal{P}(\alpha, \Xi, \{g_t\})$ for each choice of Ξ and α are shown. (b) and (c) The bank failure rates and the cross-sectional average of banks' loan growth rates, respectively, at the stationary distributions of the different (Ξ, α) combinations are shown.

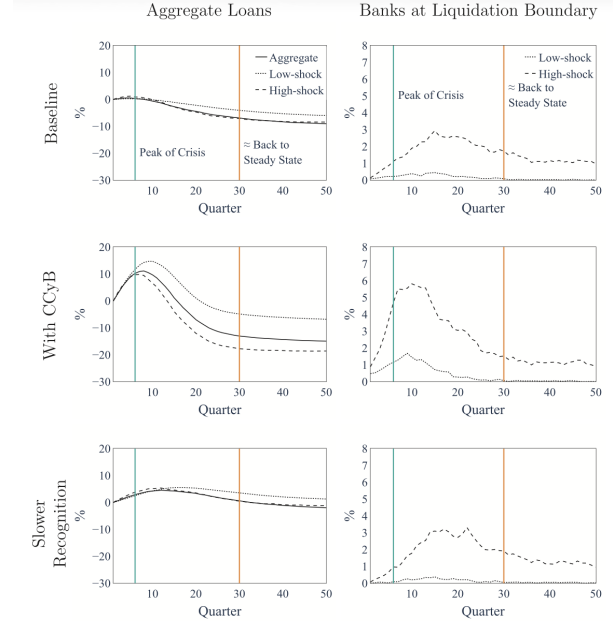


FIGURE 15
Macprudential policy effects
Notes: The figure depicts aggregate lending (left column) and the percentage share of banks at the liquidation boundary (right column) during a simulated credit crisis (as given by equation (20)) and its aftermath under three macroprudential policies. Aggregate lending is shown as a percentage deviation relative to the preceding boom's peak, normalized at $t = 0$, and detrended using the steady-state growth rate. The top row presents the baseline case, where Ξ and α are constant at their calibrated and estimated values, respectively. The middle row presents the case under CCyB, where the capital requirement, Ξ_t^{CCyB} , follows equation (21). The bottom row shows the outcome when the accounting rule α_t^{CCyB} is time-varying and relaxed after an adverse aggregate shock. The plot shows aggregate (fundamental) loans as solid lines, "low-shock" banks as dotted lines, and "high-shock" banks as dashed lines.

5.16 Figure 15: Macprudential policy effects

Read:

- Under the baseline, a credit crisis slowly reduces aggregate lending because more banks move toward liquidation.
- Under CCyB, lending initially rises more, but failures rise sharply and long-run lending ends up lower. The policy backfires.
- Under slower recognition during the crisis, the response is smoother and more targeted: high-shock banks get breathing room without creating the same ex ante leverage incentives for all banks.
- This is one of the paper's most original policy results.

6 Short summary

- **Conclusion.** Delayed loan-loss recognition is a quantitatively important source of the gap between book and market measures of bank capitalization and leverage.
- **Intuition.** Hidden losses create zombie loans, which make banks look safer on their books than they are in fundamental terms. That both delays deleveraging and weakens the link between regulatory capital and true fragility.
- **Empirical lesson.** The persistence of liabilities and leverage after excess-return shocks is informative about accounting speed, not just about generic balance-sheet adjustment costs.
- **Policy lesson.** Optimal regulation should jointly choose capital requirements and accounting rules. Faster recognition can justify looser book capital requirements.

7 Conclusion

7.1 Core conclusion

- **Main finding.** A dynamic bank model with delayed accounting can jointly match four hard-to-reconcile facts about banks' Tobin's Q , leverage distributions, and post-shock adjustment dynamics.
- **Why the mechanism works.** The delay in book recognition creates a stock of hidden losses Z that drives a wedge between fundamental and book equity and thereby changes the effective tightness of regulation.
- **Why capital requirements are second-best.** Because regulatory enforcement itself liquidates banks, capital requirements induce banks to hold buffers and cannot implement the first-best leverage choice.
- **Why accounting rules matter.** The loss-recognition speed α changes the cross-sectional distribution of hidden fragility and therefore changes how much capital buffering is privately optimal.
- **Quantitative result.** The estimated optimum substantially speeds recognition (α rises from 4.34% to about 23.6%) while relaxing the capital requirement from about 8% to roughly 5.4%.
- **Macroprudential result.** A CCyB can amplify fragility under delayed accounting, whereas temporary accounting forbearance is better targeted because it primarily affects banks that have already been hit by losses.

7.2 Economic intuition

- Book values are backward-looking; market values are forward-looking. When regulation is based on the former, there is room for hidden leverage.
- Delayed accounting has both a benefit and a cost. It softens deleveraging for banks that have been hit by losses, but it also lets banks with hidden losses remain highly levered for longer.

- Faster recognition makes the banking system safer not mainly by mechanically lowering book leverage, but by shrinking the stock of zombie loans and moving risky banks away from the liquidation boundary.
- Relaxing capital requirements can be welfare-improving once recognition is faster because book leverage becomes a better proxy for true leverage.

7.3 Takeaway

This paper is best read as a *Q-theory of bank capital under accounting frictions*. Its big message is that market and book measures of bank health are not interchangeable, and any serious model of bank regulation has to keep both in view. The most important insight is not simply that accounting delay matters, but that it changes what the optimal capital rule looks like: better accounting can substitute for stricter book-capital requirements. On the macroprudential side, the paper also delivers a sharp warning that broad-based capital relaxations may create the wrong incentives when losses are slow to show up on the books.